AEKANK PATEL

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Education

Stevens Institute of Technology Hoboken, NJ
Master of Science in Data Science, **GPA: 3.97/4.0** Aug 2024 – May 2026
Relevant Coursework: Deep Learning, Applied Machine Learning, Big Data Technologies, Numerical Linear Algebra for Big Data, Statistical Methods, Intro to Probability Theory, Deep Reinforcement Learning for Financial Applications, Time Series Analysis I

Manipal Institute of Technology Manipal, India
Bachelor of Technology in Mechatronics Engineering, **GPA: 7.83/10** Oct 2020 – Jul 2024
Relevant Coursework: Data Structures and Algorithms, Machine Vision and Image Processing, IIoT Lab

Skills

Programming: Python, R, SQL, C
Machine Learning & AI: Supervised Learning, Unsupervised Learning, Model Evaluation, Hyperparameter Tuning, Ensemble Learning, CNN, RNN, Transfer Learning, NLP, LLMs, Computer Vision, RAG, Explainable AI
Frameworks & Libraries: NumPy, Pandas, Scikit-learn, SciPy, Matplotlib, TensorFlow, Keras, PyTorch, XGBoost, OpenCV, LangChain
Data, Backend & Big Data: Flask, REST APIs, SQLite, PostgreSQL, MongoDB, Apache Spark, Hadoop, YARN
Tools: Streamlit, Git, Tableau, Power BI, Bloomberg, LSEG, Capital IQ, Groq

Experience

Stevens Institute of Technology Hoboken, NJ
Graduate Teaching Assistant Sep 2025 – May 2026
- Evaluated quantitative and Python-based assignments for 89 graduate students across Numerical Linear Algebra for Big Data, Intro to Probability Theory, and Foundational Mathematics for Data Science, delivering consistent and timely feedback.
- Standardized grading criteria by identifying recurring analytical errors and improving consistency in evaluation and feedback.

Matrix ComSec Pvt. Ltd. Vadodara, India
Research and Development Intern Jan 2024 – May 2024
- Developed a real-time human fall detection system using 100,000+ images, achieving 91.39% test accuracy after benchmarking CNN, RNN, BodyPix, and R-CNN models and deploying a CNN–MediaPipe pipeline for industrial use.
- Improved multi-person detection robustness by integrating YOLOv5, supporting deployment in dynamic industrial video environments.

Projects

NLIP Sentinel: AI Trust Firewall for Multi-Agent Workflows [Live Demo] [GitHub] May 2026
- Built a production-grade trust firewall for multi-agent AI workflows using FastAPI and React, validating agent messages, tool calls, permissions, and risky claims through a custom policy engine with a 100-point trust scoring system and full audit logging for responsible AI compliance.
- Designed a six-agent pipeline with human approval gates, sandboxed Python execution, fallback logic, retry handling, and escalation paths, demonstrating proficiency in agent-friendly system design and AI-native engineering.

Patient No-Show Risk Dashboard [Tableau] [GitHub] Mar 2026 – Apr 2026
- Built an end-to-end patient no-show prediction pipeline on 110K+ appointments by cleaning inconsistent records, designing a normalized SQLite database, and performing SQL-based analysis of appointment no-show patterns.
- Trained a Random Forest model (AUC: 0.966) with engineered behavioral and scheduling features, and created interactive dashboard to flag high-risk appointments for outreach and operational follow-up.

E-Commerce Product Search and Recommendation System Oct 2025 – Dec 2025
- Built an end-to-end e-commerce product search and recommendation pipeline on Google Cloud Platform using Apache Spark, Dataproc, and Google Cloud Storage to process 9.8M MARQO-GS-10M query-product records.
- Implemented TF-IDF, LSH-based semantic retrieval, and FP-Growth market basket analysis in Spark MLlib, achieving Precision@K of 0.782, NDCG of 0.856, MRR of 0.790, and MAP of 0.748.

FinRAG: Financial Document Intelligence System [Live Demo] [GitHub] Jun 2025 – Aug 2025
- Constructed a RAG pipeline over 25+ financial documents using HuggingFace embeddings, vector similarity search, and Groq LLaMA for finance document question answering.
- Obtained 80% retrieval accuracy through metadata filtering, prompt optimization, and confidence scoring; launched the system on Streamlit Cloud.

FRAUDGEN: Unmasking Fraud with Real-Time Explanations [Live Demo] [GitHub] Mar 2025 – May 2025
- Achieved 96.55% recall and 0.9995 AUC on the PaySim dataset (6.36M transactions, 0.1291% fraud) using XGBoost with engineered behavioral, transactional, and risk-based features.
- Validated model performance using 5-fold stratified temporal splits to address class imbalance and temporal leakage; implemented a real-time system integrating IP geolocation and VPN detection.

Certifications

Google Data Analytics, IBM AI Engineering, AWS Cloud Foundations, AWS Data Engineering, Bloomberg Market Concepts